

TUTORIAL 2: DIFFERENTIATION

1. Use definition of derivative to find $f'(x)$ for the following functions:

- a) $f(x) = 5 - 4x + 3x^2$
- b) $f(x) = 5\sqrt{x}$
- c) $f(x) = \frac{4}{2x+1}$
- d) $f(x) = \frac{x^2}{x-2}$
- e) $f(x) = 3x - \frac{1}{x}$

2. Find $\frac{dy}{dx}$ for the following functions:

- a) $f(x) = x^4(1 - 3x)^6$
- b) $f(x) = \frac{5}{\sqrt{1-x^2}}$
- c) $f(x) = \sin[\ln(x^2 - 3x)]$
- d) $f(x) = xe^{2x} + 5x$
- e) $f(x) = \sin^2(2x + 1)$
- f) $f(x) = \frac{x^2}{\sqrt{2x+3}}$
- g) $f(x) = \frac{\sin x}{1+\cos x}$
- h) $f(x) = e^{x^3-9x+7}$
- i) $f(x) = \tan(x^2 + \ln 4x^3)$
- j) $f(x) = \frac{\cos 2x}{5+\sin 2x}$
- k) $f(x) = \ln\left(\frac{5x}{1-x^2}\right)$
- l) $f(x) = \sqrt[3]{x^2 - 2}$
- m) $f(x) = \ln \frac{3x-2}{\cos^2 2x}$
- n) $f(x) = \ln \frac{e^{2x-3}}{1-4x}$
- o) $f(x) = x^4 \ln(3x - 1)$
- p) $f(x) = \frac{1}{x^2} + 2x^3 - \sqrt[3]{x}$
- q) $f(x) = \ln^2(3x + e^{2x})$

3. Find the second derivative for the following functions:

- a) $y = 5 \sin 2x + 3e^{-2x} - \frac{1}{x}$
- b) $y = \ln(5 + \sin x)$
- c) $y = \frac{e^{2x}}{1-x}$
- d) $y = \ln(x^3 - 2)$

4. If $y = e^{4x^2}$, show that $\frac{1}{4}y'' - 2xy' - 2y = 0$

5. If $y = \frac{1}{\sqrt{2x^2+1}}$, show that $y'' - y' + 2y^5 = \frac{8x^2}{(\sqrt{2x^2+1})^5} + \frac{2x}{(\sqrt{2x^2+1})^3}$

6. Find $\frac{dy}{dx}$ using the implicit differentiation
- $5ye^{1-x} + 5\cos 2y = x^3 + 7$
 - $y^2e^{5x} + \cos(xy) = 2 + \ln(x + 1)$
 - $\ln(3 - 2e^{1-x}) + x^2y + 3 = \frac{3x}{x+2} + 2y$
 - $e^{3xy} - \cos y = \ln(2x + 3)$

7. Find $\frac{dy}{dx}$ using the parametric differentiation
- $x = t^2 + 1, y = t^3 - 1$
 - $x = 3\cos t, y = 3\sin t$
 - $x = t + \sqrt{t}, y = t - \sqrt{t}$
 - $x = 2t^3 + 1, y = t^2 \cos t$
 - $x = te^{-t}, y = 2t^2 + 1$

8. Find $\frac{d^2y}{dx^2}$ using the parametric differentiation

- $x = \sin t, y = \cos t$
- $x = 3t^2 + 1, y = t^3 - 2t^2$
- $x = \frac{1}{2}t^2 + 2, y = \sin(t + 1)$
- $x = e^{-t}, y = t^3 + t + 1$
- $x = 3t^2 + 4t, y = \sin 2t$

Answer:

- $-4 + 6x$
 - $\frac{5}{2\sqrt{x}}$
 - $-\frac{8}{(2x+1)^2}$
 - $\frac{x^2-4x}{(x-2)^2}$
 - $3 + \frac{1}{x^2}$
- $2x^3(1-3x)^5(2-15x)$
 - $\frac{5x}{(1-x^2)^{\frac{3}{2}}}$
 - $\frac{(2x-3)\cos[\ln(x^2-3x)]}{x^2-3x}$
 - $2xe^{2x} + e^{2x} + 5$
 - $4\sin(2x+1)\cos(2x+1)$
 - $\frac{3x^2+6x}{(2x+3)^{\frac{3}{2}}}$
 - $\frac{1}{1+\cos x}$
 - $(3x^2-9)e^{x^3-9x+7}$
 - $\frac{(2x^2+3)\sec^2[x^2+\ln(4x^3)]}{x}$
 - $\frac{-10\sin 2x-2}{(5+\sin 2x)^2}$
 - $\frac{1}{x} + \frac{2x}{1-x^2}$
 - $\frac{2x}{3(x^2-2)^{\frac{2}{3}}}$
 - $\frac{3}{3x-2} + 4\tan 2x$
 - $\frac{2e^{2x}}{e^{2x}-3} + \frac{4}{1-4x}$
 - $\frac{3x^4}{3x-1} + 4x^3\ln(3x-1)$
 - $-\frac{2}{x^3} + 6x^2 - \frac{1}{3x^{\frac{2}{3}}}$
 - $\frac{2(3+2e^{2x})\ln(3x+e^{2x})}{3x+e^{2x}}$
- $-20\sin 2x + 12e^{-2x} - \frac{2}{x^3}$
 - $\frac{-5\sin x-1}{(5+\sin x)^2}$
 - $\frac{e^{2x}(4x^2-12x+10)}{(1-x)^3}$
 - $\frac{-3x^4-12x}{(x^3-2)^2}$
- DIY
- DIY

6. a) $\frac{3x+5ye^{1-x}}{5e^{1-x}-16\sin 2y}$ b) $\frac{\frac{1}{x+1}-5y^2e^{5x}+y\sin(xy)}{2ye^{5x}-x\sin(xy)}$ c) $\frac{\frac{2e^{1-x}}{3-2e^{1-x}}+2xy-\frac{6}{(x+2)^2}}{2-x^2}$ d) $\frac{\frac{2}{2x+3}-3ye^{3xy}}{3xe^{3xy}+\sin y}$

7. 1. a) $\frac{3t}{2}$ b) $-\cot t$ c) $\frac{2\sqrt{t}-1}{2\sqrt{t}+1}$ d) $\frac{2\cos t-t\sin t}{6t}$ e) $\frac{4te^t}{1-t}$

8. 1. $-\sec^3 t$ 2. $\frac{1}{12t}$ 3. $\frac{-t\sin(t+1)-\cos(t+1)}{t^3}$
4. $(3t^2+6t+1)e^{2t}$ 5. $\frac{-2(3t+2)\sin 2t-3\cos 3t}{2(3t+2)^3}$